(Start date: 03:05:45 PM, 02 August 2021)

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$

Select the correct option

- $I = \frac{5M}{2a^3}$
- $I = \frac{2}{5}Ma^2$

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Question # 10 of 10 (Start time: 11:56:53 AM, 02 August 2021)

If a rigid body is rotating about a fixed point on ground then its kinetic energy T is defined by

Select the correct option

○	$T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2$
○	$T = \frac{1}{2}I_0\omega^2$

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Question # 10 of 10 (Start time: 03:06:27 PM, 02 August 2021)

The radius of gyration K of the hemisphere of mass M and moment of inertia $I = \frac{2}{5}Ma^2$ about an axis through its center is

Select the correct option

- $K = \sqrt{\frac{2}{5}}a$
- $K = \frac{1}{a}\sqrt{\frac{5}{2}}$

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Question # 7 of 10 (Start time: 03:05:00 PM, 02 August 2021)

Kinetic energy T of a rigid body in a planar motion, is defined by

Select the correct option

- $T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2$
- $T = \frac{1}{2}I_0\omega^2$

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Question # 3 of 10 (Start time: 11:50:16 AM, 02 August 2021)

For an elastic body, the distance between two particles.....under the action of applied force

Select the correct option

- remains constant
- is changed

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Question # 1 of 10 (Start time: 02:51:53 PM, 02 August 202

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow .....

Select the correct option

- a straight line
- a parabola

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Question # 8 of 10 (Start time: 11:54:51 AM, 02 August 2021)

If P is any point on the ellipsoid of inertia, then the moment of inertia of a rigid body about any line $\vec{OP}$ is equals to the reciprocal of the square of the length of $|\vec{OP}|$

Select the correct option

○	true
○	false

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Question # 9 of 10 (Start time: 02:57:28 PM, 02 August 2021)

If a body rotates about some external fixed point it is called revolution

Select the correct option

- spin
- revolution

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Question # 6 of 10 (Start time: 11:53:05 AM, 02 August 2021)

The position vector of the center of mass of a system of particles is

Select the correct option

- $\vec{r}_c = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$
- $\vec{r}_c = \frac{\sum_i m_i \vec{r}_i}{\sum_i \vec{r}_i}$

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MTH622:Quiz 3

Question # 8 of 10 (Start time: 02:55:51 PM, 02 August 2021)

Consider a right circular cone of density ρ, radius a and height h is composed of elementary circular discs consider a typical disc of radius r and width δz them mass of the disc is

Select the correct option

- $\delta m = \rho 2\pi r \delta z$
- $\delta m = \rho \pi r^2 \delta z$

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Question # 4 of 10 (Start time: 02:53:05 PM, 02 August 2021)

Dynamics deals with the forces acting on any object and relationship fundamentally to the motion.

Select the correct option

- true
- false

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Question # 7 of 10 (Start time: 02:55:16 PM, 02 August 2021)

Angular momentum can be defined as

Select the correct option

- $L = p \times \omega$
- $L = r \times v$
- $L = r \times \omega$
- $L = r \times p$

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Question # 7 of 10 (Start time: 11:53:57 AM, 02 August 2021)

If angular momentum $\Omega = 2\omega \sin \omega t$ then the torque τ is

derivative of Ω

Select the correct option

- $-2 \cos \omega t$
- $2\omega^2 \cos \omega t$ Ans

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Quiz Start

Question # 5 of 10 (Start time: 03:03:19 PM, 02 August 2021)

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass m with velocity $3\hat{i}$. If after collision the velocity of first particle is $5\hat{i}$ and that of second particle is $2\hat{i}$ find m

Select the correct option

- 4
- 3

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MTH622:Quiz 3

Question # 1 of 10 (Start time: 11:48:22 AM, 02 August 2021)

The torque acting on a particle is proportional to the time rate of change in its angular acceleration.

Select the correct option

- false
- true

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Question # 4 of 10 (Start time: 11:50:46 AM, 02 August 2021)

In general, the directions of angular velocity and angular momentum are

Select the correct option

- distinct
- same

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Question # 9 of 10 (Start time: 11:55:39 AM, 02 August 2021)

If the axis of rotation passes through
the center of mass of the rigid body then it is called

Select the correct option

- spin
- revolution

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Question # 10 of 10 (Start time: 02:57:41 PM, 02 August 2

If an object explodes, the center of mass will

Select the correct option

- come into rest
- continue its previous motion

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$m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$

$(4)(4\hat{i}) + m_2(3\hat{i}) = (4)(5\hat{i}) + m_2(2\hat{i})$

$16\hat{i} + 3m_2\hat{i} = 20\hat{i} + 2m_2\hat{i}$

$3m_2\hat{i} - 2m_2\hat{i} = 20\hat{i} - 16\hat{i}$

$m_2\hat{i} = 4\hat{i}$

$\boxed{m_2 = 4\,kg}$

$m_1 = 4 \quad v_1 = 4\hat{i} \quad m_2 = 5 \quad v_2$

$u_1 = ? \quad u_2 = 2\hat{i}$

$m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$

$(4)(4\hat{i}) + (5)(3\hat{i}) = (4)(u_1) + (5)(2\hat{i})$

$16\hat{i} + 15\hat{i} = 4u_1 + 10\hat{i}$

$4u_1 = 21\hat{i}$

$\boxed{u_1 = \frac{21}{4}\hat{i}}$

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